



NEVER STOP

Tokens, Constants, Variables & Data Types

Unit 3 · Lecture 2

ICSE Class 9 · Computer Science · Java

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SECTION 1 : TOKENS IN JAVA

What is a Token?

A **Token** is the *smallest individual unit* of a Java program that is meaningful to the compiler. Just like a sentence is made of words, {

Real-Life Analogy

Think of a Java program as a sentence.

Just as a sentence has words, punctuation, and spaces — a Java program has tokens: keywords, identifiers, operators, etc.

Types of Tokens

Token Type	Description	Examples
Keyword	Reserved words with special meaning	int, float, class, if, while
Identifier	Names given by the programmer	age, name, myVar, Sum
Literal	Fixed values written directly in code	42, 3.14, 'A', "Hello"
Operator	Symbols performing operations	+, -, *, /, %, =, ==
Separator	Punctuation marks used to structure code	{ } () [] ; ,

Keywords

Java has **53 reserved keywords**. They cannot be used as variable names or identifiers. All keywords are written in **lowercase**.

<code>abstract</code>	<code>assert</code>	<code>boolean</code>	<code>break</code>	<code>byte</code>	<code>case</code>
<code>catch</code>	<code>char</code>	<code>class</code>	<code>const*</code>	<code>continue</code>	<code>default</code>
<code>do</code>	<code>double</code>	<code>else</code>	<code>enum</code>	<code>extends</code>	<code>final</code>
<code>finally</code>	<code>float</code>	<code>for</code>	<code>goto*</code>	<code>if</code>	<code>implements</code>
<code>import</code>	<code>instanceof</code>	<code>int</code>	<code>interface</code>	<code>long</code>	<code>native</code>
<code>new</code>	<code>null*</code>	<code>package</code>	<code>private</code>	<code>protected</code>	<code>public</code>
<code>return</code>	<code>short</code>	<code>static</code>	<code>strictfp</code>	<code>super</code>	<code>switch</code>
<code>synchronized</code>	<code>this</code>	<code>throw</code>	<code>throws</code>	<code>transient</code>	<code>try</code>
<code>var</code>	<code>void</code>	<code>volatile</code>	<code>while</code>		

* `const` and `goto` are reserved but unused. `null` is a literal.

Rules for Identifiers

Rule	Detail	
■ Can start with	Letter (A-Z, a-z), underscore (_), or dollar sign (\$)	
■ Can contain	Letters, digits (0-9), _ , \$	
Rule	Detail	
■ Cannot start with	A digit (0-9)	
■ Cannot use	Spaces or special characters like @, #, !	
■ Cannot be	A Java keyword (int, class, etc.)	
■■ Case sensitive	age, Age, AGE are THREE different identifiers	
■■ No length limit	But keep names meaningful and readable	
Valid ■	Invalid ■	Reason (for invalid)
myAge	1age	Starts with digit
_name	my name	Contains space
\$price	class	Reserved keyword
totalSum	my@var	Special character @

■ SECTION 2 : CONSTANTS IN JAVA

What is a Constant?

A **constant** is a value that *does not change* during the execution of a program. Once assigned, it stays fixed forever.

■ Real-Life Analogy

The value of π (pi) = 3.14159... never changes → it is a constant!

The speed of light = 3×10^8 m/s → also a constant in physics.

Types of Constants in Java

Type	Description	Example
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Integer Constant	Whole numbers (no decimal)	10, -5, 0, 255
Float/Double Constant	Numbers with decimal point	3.14, -0.5, 2.718
Character Constant	Single character in single quotes	'A', 'z', '9', '#'
String Constant	Text in double quotes	"Hello", "Java"
Boolean Constant	Only two values: true or false	true, false

Declaring Constants in Java using final

Use the **final** keyword to declare a constant. By convention, constant names are written in ALL_CAPS.

```
final int MAX_MARKS = 100;
final double PI = 3.14159;
final char GRADE = 'A';

final String SCHOOL = "NeverStop Academy";
final boolean IS_PASS = true;

// Trying to change a final variable gives COMPILE ERROR:
// MAX_MARKS = 90; ■ Error: cannot assign a value to final variable
```

■ SECTION 3 : VARIABLES IN JAVA

What is a Variable?

A **variable** is a named memory location that stores a value which *can change* during program execution. Think of it as a labelled box that holds data.

■ Real-Life Analogy

A variable is like a labelled jar in your kitchen.

The label (name) stays the same, but you can put different things inside (values).

e.g. jar labeled "sugar" can be refilled → `int sugar = 500; sugar = 300;`

Syntax for Declaring a Variable

```

dataType variableName; // Declaration only
dataType variableName = value; // Declaration + Initialization

// Examples:
int age; // Declare age
= 15; // Assign later
int marks = 95; // Declare and assign
together double temperature = 36.6; char
grade = 'A'; String name = "Rahul"; boolean
isPassed = true;

```

Variable vs Constant — Quick Comparison

Feature	Variable	Constant
Value	Can change	Cannot change
Keyword	No special keyword	final keyword
Naming Style	camelCase (e.g. myAge)	ALL_CAPS (e.g. MAX_SIZE)
Memory	Value can be overwritten	Memory value is locked
Example	int score = 50; score = 80;	final int MAX = 100;

Types of Variables in Java

Type	Where Declared	Scope / Life
Local Variable	Inside a method or block	Exists only within that block
Instance Variable	Inside class, outside methods	Exists as long as the object lives
Class Variable (static)	Inside class with 'static' keyword	Shared by all objects of the class

SECTION 4 : DATA TYPES IN JAVA

What is a Data Type?

A **data type** tells the compiler *what kind of data* a variable will hold, and *how much memory* to allocate for it. Java is a **strongly-typed** language — every variable must have a declared type.

Real-Life Analogy

Think of data types as different sized boxes:

A small box for a single letter (char), a medium box for a whole number (int), and a large box for a very precise decimal (double).

Classification of Data Types

Category	Types	Description
Primitive (Built-in)	byte, short, int, long, float, double, char, boolean	Basic data types — directly store values in memory
Non-Primitive (Reference)	String, Arrays, Classes, Interfaces	Store memory addresses (references) to objects

Primitive Data Types — Detailed Table

Type	Size	Range	Default	Example
byte	1 byte (8 bits)	-128 to 127	0	byte b = 100;
short	2 bytes (16 bits)	-32,768 to 32,767	0	short s = 500;
int	4 bytes (32 bits)	-2,147,483,648 to 2,147,483,647	0	int i = 1000;
long	8 bytes (64 bits)	-9.2×10^{18} to 9.2×10^{18}	0L	long l = 9999L;
float	4 bytes (32 bits)	$\sim \pm 3.4 \times 10^{38}$ (7 decimal digits)	0.0f	float f = 3.14f;
double	8 bytes (64 bits)	$\sim \pm 1.8 \times 10^{308}$ (15 decimal digits)	0.0d	double d = 3.14;
char	2 bytes (16 bits)	0 to 65,535 (Unicode)	'\u0000'	char c = 'A';

boolean	1 bit (JVM specific)	true or false only	false	boolean b = true;
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Memory Size Order (Smallest → Largest)

byte 1B	short 2B	int 4B	long 8B	float 4B	double 8B
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Note: float and double are for decimals; byte/short/int/long are for whole numbers.

Type Casting (Conversion)

Converting one data type to another is called **type casting**.

Type	Direction	Done By	Example
Widening (Implicit)	smaller → bigger type	Automatically by Java	int i=5; double d=i;
Narrowing (Explicit)	bigger → smaller type	Manually by programmer	double d=5.9; int i=(int)d; // i=5

■ SECTION 5 : COMPLETE JAVA EXAMPLE

```
// ■■ Complete Example: Tokens, Constants, Variables & Data Types ■■

public class StudentInfo {    // Instance variables
    (different data types)    String name = "Rahul Sharma";
    // String (non-primitive)    int age = 15;
    // int (primitive)          double percentage = 87.5;        //
    double (primitive)          char grade = 'A';                //
    char (primitive)            boolean isPassed = true;         //
    boolean (primitive)

    // Constant (final keyword, ALL_CAPS
    naming)    final int MAX_MARKS = 100;
    final double PI = 3.14159;

    public static void main(String[] args) {
        StudentInfo s = new StudentInfo();

        // Displaying values
        System.out.println("Name      : " + s.name);
        System.out.println("Age       : " + s.age);
        System.out.println("Percentage: " + s.percentage + "%");
        System.out.println("Grade     : " + s.grade);
        System.out.println("Passed?   : " + s.isPassed);
        System.out.println("Max Marks : " + s.MAX_MARKS);

        // Type Casting: double → int (Narrowing)        int
        approxMarks = (int)(s.percentage);
        System.out.println("Approx Marks : " + approxMarks); // 87

        // Type Casting: int → double (Widening - automatic)
        double exactAge = s.age;        System.out.println("Age as double:
        " + exactAge);    // 15.0
    }
}
```

Output:

```
Name : Rahul Sharma
Age : 15
Percentage: 87.5%
Grade : A
Passed? : true Max Marks : 100 Approx Marks : 87
Age as double: 15.0
```


■ SECTION 6 : MEMORY TRICKS & MNEMONICS

1■■■ Token Types → 'KILOS'

K = Keywords | I = Identifiers | L = Literals | O = Operators | S = Separators

■ *"Every Java Program KILOS (Kills) errors when tokenised!"*

2■■■ Primitive Types → 'B S I L F D C B' byte → Short → Int → Long → Float →

Double → Char → Boolean

■ *"Big Strong Indian Lions Fight During Cold Blizzards"*

3■■■ final = Frozen

The final keyword FREEZES the value — just like ice freezes water.

■ *final int X = 5; → Think: X is frozen forever at 5.*

4■■■ Variable = Jar with a Label

The jar label = variable name. What's inside = variable value.

■ *You can empty the jar and refill it → that's changing a variable's value!*

5■■■ Widening = Wide road (easy to go from small car to big truck lane) byte → short

→ int → long → float → double (no data loss)

■ *Narrowing = Narrow lane (big truck into small lane — some data may fall off!)*

6■■■ int vs double int → No decimal point. double → Has a decimal point.

■ *"int = Integer. double = Decimal (double precision)."*

7■■■ char uses single quotes, String uses double quotes

char c = 'A'; (one character = one set of single quotes)

■ *String s = "Hello"; (multiple characters = double quotes)*

■ SECTION 7 : PRACTICE QUESTIONS

Part A — Multiple Choice Questions (MCQ)

1. Which of the following is a valid Java identifier?

- (a) 2name
- (b) _name
- (c) int
- (d) my name

Answer: (b) _name

2. Which keyword is used to declare a constant in Java?

- (a) const
- (b) static
- (c) final
- (d) fixed

Answer: (c) final

3. What is the size of a double data type in Java?

- (a) 2 bytes
- (b) 4 bytes
- (c) 6 bytes
- (d) 8 bytes

Answer: (d) 8 bytes

4. Which of the following is NOT a primitive data type in Java?

- (a) int
- (b) float
- (c) String
- (d) boolean

Answer: (c) String

5. What will be the output of: `int x = (int)9.99; System.out.println(x);`

- (a) 9.99
- (b) 10
- (c) 9
- (d) Error

Answer: (c) 9 (narrowing removes decimal part)

Part B — Short Answer Questions

1. Differentiate between a variable and a constant with suitable Java examples.

Answer:

2. List any 5 rules for naming identifiers in Java.

Answer:

3. Write a Java program to declare one variable of each primitive data type and print its value.

Answer:

4. What is the difference between int and long? When would you use long?

Answer:

5. What are tokens? List all types of tokens with one example each.

Answer:

6. What is type casting? Distinguish between widening and narrowing conversion.

Answer:

7. What will happen if you try to assign a new value to a final variable? Give an example.

Answer:

Part C — Coding Challenges ■

Easy ■

Write a Java program that stores your name, age, class, and percentage. Print all of them using `System.out.println()`.

Your Code Here:

Medium ■■

Declare a constant for the value of PI (3.14159). Then compute and print the area of a circle with radius 7. Formula: $\text{area} = \text{PI} \times r \times r$

Your Code Here:

Hard ■■■

Write a Java program that demonstrates: (a) widening type conversion from int to double, and (b) narrowing type conversion from double to int. Print the values before and after conversion and observe the difference.

Your Code Here:

■ QUICK REVISION CARD — At a Glance

Concept	Key Point	Example
Token	Smallest unit of a Java program (5 types: KILOS)	int, age, +, ;
Keyword	53 reserved words, all lowercase	int, class, if, while
Identifier	Programmer-given name; can't start with digit	myAge, _total
Literal	Fixed value written in code	42, 3.14, 'A', true
Constant	Value that never changes; use final keyword	final int MAX=100;
Variable	Named memory location whose value can change	int age = 15;
int	4 bytes whole numbers most common type	int x = 5;
double	8 bytes decimal numbers high precision	double d = 3.14;
char	2 bytes single character in single quotes	char c = 'A';
boolean	1 bit only true or false	boolean b = true;
Widening	small → big type automatic no data loss	int i=5; double d=i;
Narrowing	big → small type manual cast may lose data	int i=(int)9.9; //9

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